

Design and Performance Analysis of a Novel 8-bit Reversible ALU Using 16nm CMOS Technology

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Abstract— Reversible logic has emerged as a key component of contemporary circuit design due to the increasing demand for ultra-low-power and high-performance digital systems in nanoscale technologies. This study worked on reversible gate-based Arithmetic Logic Units (ALUs) and conventional ALUs. Reversible base ALU works on different types of reversible gates, such as Feynman, URG, Fredkin, and Peres. This paper proposed that a novel 8-bit reversible ALU performed with a 21.29% delay, 69.38% power, and 66.38% energy improvement noticed when compared with conventional reversible ALU. Additionally, the novel 8-bit reversible ALU improves delay, power, and energy by 26.69%, 69.74%, and 67.87%, respectively, compared to the existing CMOS ALU. All performances are analyzed using 16 nm Complementary Metal Oxide Semiconductor (CMOS) technology, and the design is synthesized. This assessment employed Cadence Virtuoso software for simulations and meticulous data analysis.

Keywords— CMOS, Reversible Logic, ALU, Energy, Power consumption, Delay.

I. INTRODUCTION

As of nowadays, with persistent computing and mobile devices, power consumption has emerged as an indispensable design consideration in digital circuit. In the semiconductor industry, reversible logic gates are vital because it reduce noise and power consumption, the two parameters that are imperative for power-efficient architecture. Reversible computing is a paradigm of computation that enables the reversal of computational procedures across time, thereby restoration of prior system states [1]. ALUs are the computational core of a processor, responsible for performing arithmetic and logical operations. Furthermore, as a vital component, the ALU has a substantial impact on the overall energy efficiency of the digital processor. Among the most crucial parts of a Central Processing Unit (CPU) is the ALU [2]. As there is a demand for lower power consumption, and the past few decades, reversible logic has played a major role in low-power Very-Large-Scale Integration (VLSI) circuit design methodologies. Numerous researchers have worked on reversible logic gates and proposed their methods.

In this research [3], the author tries to develop a new design along with execution for a hybrid 32-bit ALU. In this, we noticed that various reversible gate designs escalate circuit complexity, presenting a challenge that may hinder large-scale integration. This research [4] works on low power consumption and high-speed performance in an 8-bit ALU architecture employing a 17-transistor full adder. They proposed that their design can reduce power requirements for digital processor. Although, it fails to consider physical fabrication, noise margin, or temperature effects. The analysis in [5] represents that 8-bit ALU using reversible logic circuit had reduced the gate count and transistor count. For this case, propagational delay results more efficient when compared with other existing design. In addition, it observed the design retains a moderate quantity of garbage outputs, and an analysis of power or area at the layout level was not provided. This research [6] author try to develop Team model and CMOS technology of 4-bit ALU design. They proposed that their design can decrease chip area that can optimize the power consumption. The design has voltage degradation in outputs due to resistive characteristics and lacks experimental validation beyond simulations, potentially impacting real-world reliability and scalability. Reference [2] developed a power-efficient architecture for 8-bit and 16-bit reversible ALUs. It also analyzed ALU development power, size, and Look Up Table (LUT) needs. The successful approach of reversibility reduced circuit power loss while applying minimal power. Even though this design does not investigate delay optimization, noise impacts, or post-layout verification either. In [7], the author built a complex, reversible ALU using Quantum-dot Cellular Automata (QCA) technology. This paper represents that delay, energy, area, cost and power consumption performing well. The architecture is limited to single-bit operations and increases structural complexity owing to its multilayer arrangement in QCA. The analysis in [1] designing reversible QCA based ALU logic circuits in a way that is both physically and logically reversible can provide reduction in remarkable energy dissipation savings. To address the drawbacks of conventional reversible and CMOS-based ALUs, this research offers a power-efficient, functionally expanded reversible ALU.

The first contribution of this work is the implementation of an input control strategy that varies input combinations using the S0 and S1 pins, allowing a single unit to perform multiple operations. This mechanism enhances efficiency by reducing circuit complexity. Second contributions, a central control mechanism controlled by S2 and S3 signals, which ensures that only one gate is active at a time while other gates remain inactive. This strategy significantly reduced switching activity, resulting in reduced power loss and overall energy savings. The third contribution is that combining centralized control with operation selection logic enables synchronized implementation, which reduces delays and significantly improves circuit efficiency.

II. SIMULATION OF REVERSIBLE LOGIC GATES

This simulation work represents modeled key reversible gates, including the URG, Fredkin, Peres, and Feynman gates, to analyze their behavior and performance. Also include their quantum cost. For each gate, we constructed a truth table that specifies how both inputs together with outputs relate to one another, ensuring that the operations are information-conserving, as required by reversible logic. For implementation, design and circuit diagrams of each gate has employed 16nm CMOS technology in Cadence Virtuoso software. The output waveforms obtained from the simulations highlight the dynamic performance of these gates, illustrating how the outputs respond to input changes over time. This comprehensive simulation validates the correct functionality of the reversible gates and showcases their potential for energy-efficient digital circuit design.

A. Feynman Gate

This gate, referred to as CNOT (Controlled NOT) gate, is a reversible 2×2 gate. The integrated circuit equivalent of the Feynman gate is depicted in Figure 1 [7]. The system functions using two bits, referred to as qubits: one is the control bit and the other as the target bit. The gate alters the target bit exclusively when the control bit is activated; in all other cases, the emphasis bit is still unchanged.

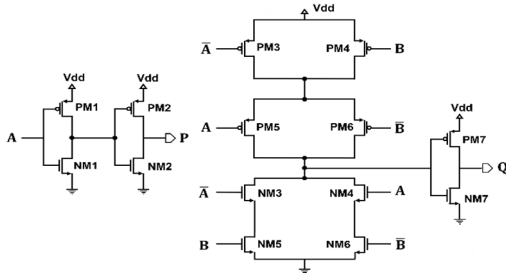


Fig. 1. Circuit diagram of Feynman gate.

The output equation is $P = A$ and $Q = A \oplus B$, while the values A as well as B represent the inputs, as opposed to P and Q are the outputs. Table I represents the truth table, and Figure 2 illustrates the operation waveform. Besides, this timing diagram and truth table have met all the criteria with each other.

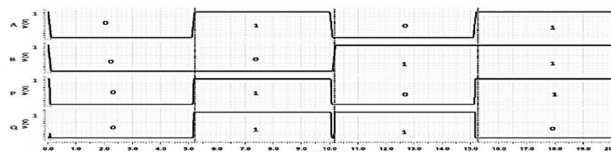


Fig. 2. Operation of Feynman gate.

TABLE I. FEYNMAN GATE OPERATION TABLE

INPUT		OUTPUT	
A	B	P	Q
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

B. Fredkin Gate

The Fredkin gate, which is formally called the Controlled-SWAP (CSWAP) gate, acts as a universal reversible logic gate commonly used in both classical reversible computing and quantum computing [8]. Figure 3 exhibits the CMOS architecture of Fredkin gate.

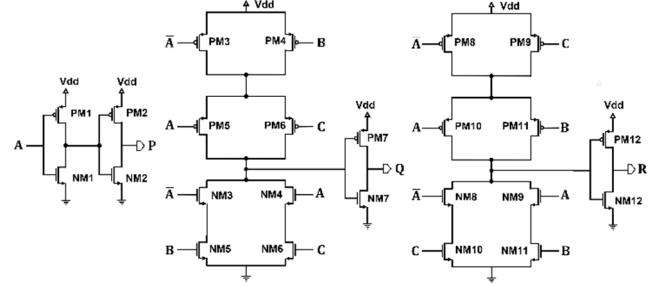


Fig. 3. Circuit diagram of Fredkin gate.

It takes three inputs as well as outputs bits. The first input remains unchanged and appears as the first output. The gate performs a conditional swap of the second and third inputs based on the value of the first input. The final output equations are $Q = \bar{A}B + AC$ and $R = \bar{A}C + AB$, where A is the control input, and B and C are the other two inputs. The CMOS implementation Fredkin gate operation waveform is depicted in Figure 4.

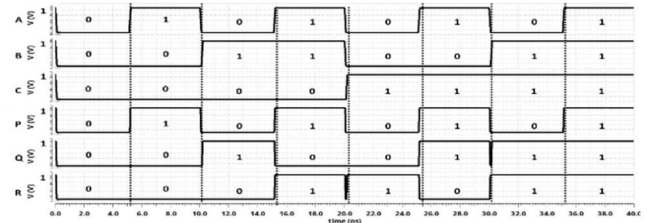


Fig. 4. Operation of Fredkin gate.

The Fredkin gate truth table is provided in Table II. Particularly A, B and C represent as input and in contrast P, Q and R working for output.

TABLE II. LOGIC TABLE OF FREDKIN GATE

Input			Output		
A	B	C	P=A	Q	R
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	0	1
1	1	1	1	1	1

C. Peres Gate

The PG referred to as the Peres gate, serves as a mechanism that turns three input values into three unique outputs, which is widely utilized in the design of reversible system [9]. This gate is particularly valued for its efficiency and low-power characteristics, making it essential in both quantum computing and classical reversible computing applications. In Figure 5 shows, CMOS implementation of Peres gate.

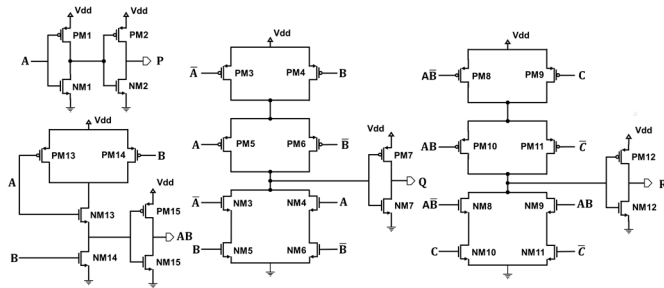


Fig. 5. Circuit diagram of Peres gate.

The first input and output remain identical. The second and third outputs are computed using the equations $Q = A \oplus B$ as well as $R = AB \oplus C$, where A, B, and C are the inputs. These logical operations allow the Peres gate to perform essential reversible computations with minimal energy dissipation. The CMOS circuit diagram output waveform is presented in Figure 6 to verify the gate's correct operation and performance characteristics.

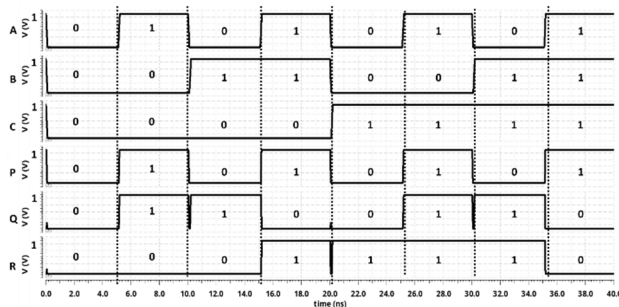


Fig. 6. Operation of Peres gate.

According to CMOS implementation of Peres gate operation truth table is provided in Table III. A, B and C working as input, as opposed to R, Q and P working as output.

TABLE III. PERES GATE FUNCTION TABLE

Input			Output		
A	B	C	$P=A$	Q	R
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

D. URG

The URG (Universal Reversible Gate) is a type of reversible logic gate constructed to execute multiple logic functions within a single gate structure [10]. Figure 7 illustrates the CMOS implementation of the URG.

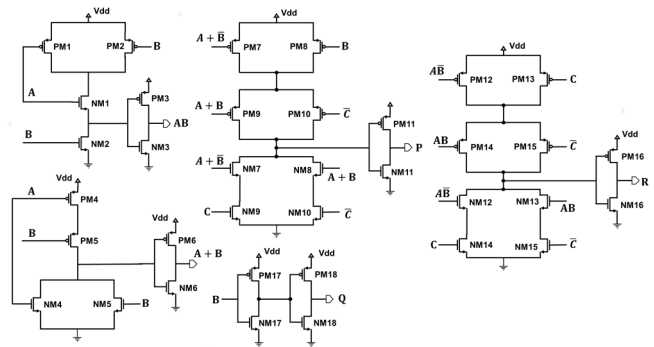


Fig. 7. Circuit diagram of URG.

The system, which is universal and allows for complex reversible circuits, can implement any Boolean function. URG maintains reversibility through multiple inputs and outputs, ensuring no data loss and synchronizing each output state with its input. Low-power, quantum, and nanotechnology computer systems that preserve energy and information are well-suited for URG. The second output remains the same as the second input. The first and third outputs are computed using the equations $P = (A+B) \oplus C$ as well as $R = AB \oplus C$, where A, B, and C are the inputs. Figure 8 demonstrates the operation of URG.

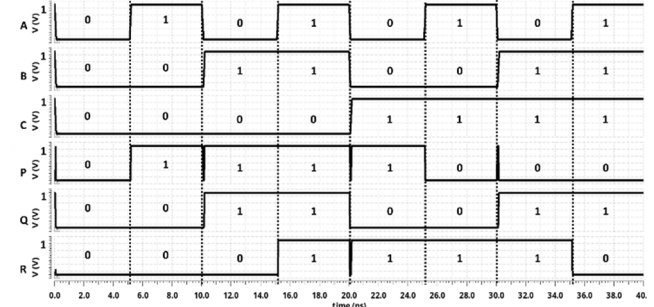


Fig. 8. Operation of URG.

Table IV. shows the truth table of URG. Here below briefly describe about URG reversible logic gate for Input and Output.

TABLE IV. URG OPERATION TABLE

Input			Output		
A	B	C	P	$Q=B$	R
0	0	0	0	0	0
0	0	1	1	0	1
0	1	0	1	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	0	0	1
1	1	0	1	1	1
1	1	1	0	1	0

E. Existing ALU

This section described both CMOS-based and reversible gate-based ALU. Figure 9 presents CMOS-based ALU that use regular logic gates like AND, OR, and XOR to perform basic math and logic tasks that processors often need [6], [11].

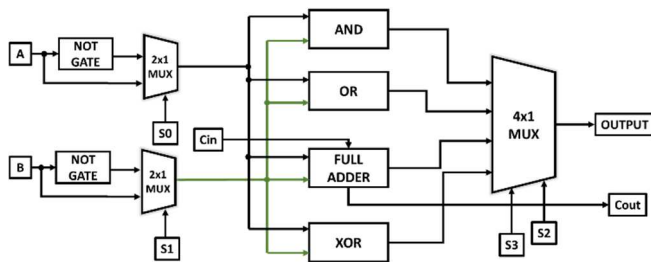


Fig. 9. Block diagram of existing CMOS based ALU.

Figure 10 shows that the reversible ALU design employs of reversible gates, such as Feynman, Peres, Fredkin, and URG, to construct reversible ALU.

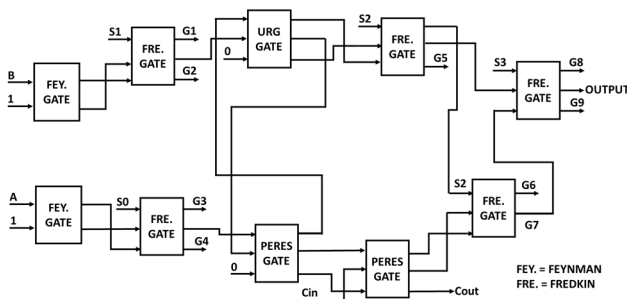


Fig. 10. Circuit diagram of existing reversible ALU.

Although reversible ALUs offer advantages in sensitivity of power applications as well as computing of quantum requirements, both CMOS and reversible systems both demonstrate a basic limitation. The main problem with current CMOS and reversible ALU is that when the control pin chooses a specific gate to give an output, other logic and arithmetic parts are still working and generating results. The simultaneous operation of all gates leads to heightened power consumption and energy loss. The proposed design resolves this issue by implementing a control logic system that activates a single gate at any given time, thereby ensuring that all other gates remain inactive.

III. PROPOSED REVERSIBLE ALU

The proposed reversible ALU is built with a combination of four important reversible logic gates (Peres, URG, Fredkin, and Feynman) and performs logical and arithmetic operations. Each reversible gate is chosen derived from its functional efficiency and minimal overhead in implementing reversible logic. Logical and arithmetic operations, including XOR, AND, NOR, NAND, OR, XNOR, addition, increment and subtraction. To maintain strict functional isolation, build a separate logic block for each operation and use the corresponding gate to implement it. In this, there are two control systems: one is an input controller, and the other is a central control unit. The input controllers are controlled by the S0 and S1 pins, which mainly work for variation on input and achieve more operations by using one gate. Conversely, the central control unit utilizes the S2 and S3 pins to activate each unit. This feature of the ALU regulates the activation of

operation blocks and serves as a key component of the design. Control signals activate the corresponding gate module for each selected operation, keeping all other modules inactive. This selective activation improves power performance by eliminating unnecessary switches while enhancing circuit efficiency. Figure 11 shown 1-bit reversible ALU has been proposed. This circuit is developed by four Feynman gates, eleven Fredkin gates, two Peres gates, and one URG gate.

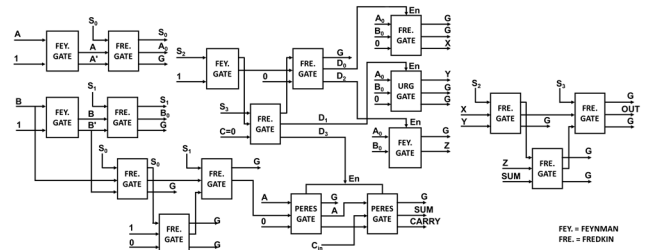


Fig. 11. Proposed 1-Bit reversible ALU.

Figure 12 presents the proposed ALU block diagram, that include an input control unit, a central control unit, and functional components such as the complete adder and logic unit. The control unit oversees data flow and operation selection, enabling an efficient performance of arithmetic and logical operations within the ALU.

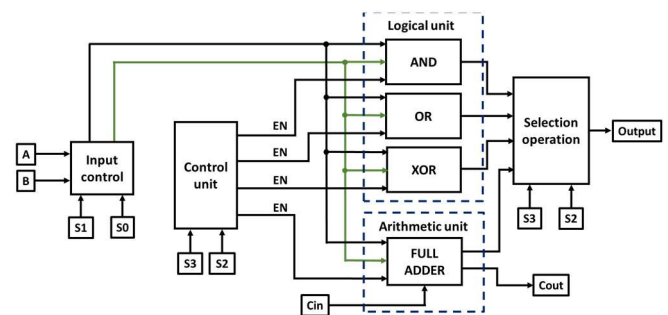


Fig. 12. Proposed reversible ALU block representation.

Figure 13 shows 8-bit reversible ALU has been proposed. The proposed 1-bit reversible ALU design has been expanded to an 8-bit configuration, and the resultant circuit was evaluated via transient analysis in Cadence Virtuoso software. An 8-bit reversible ALU consists of two main inputs: A, which includes A0 to A7, and B, which includes B0 to B7.

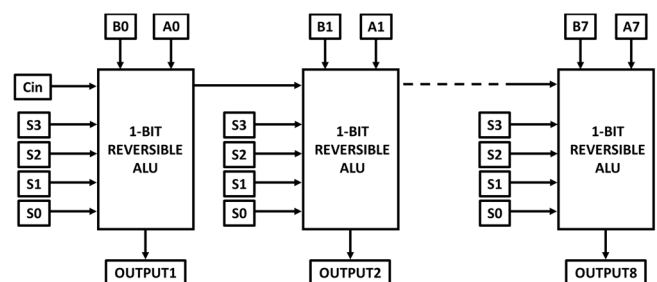


Fig. 13. An 8-bit novel reversible ALU block diagram.

Selection lines allow an n-bit ALU to recognize within operations pertaining to logic and arithmetic. If the number of selection lines is k, (such as S0, S1, S2, ..., Sk-1), the ALU performs thousands of operations. The variables A, B, and Cin are served as the inputs to the ALU. In the proposed ALU design, the four selection lines are denoted by the letters S0, S1, S2 and S3. The S3 and S2 pins are used for selecting

logical and arithmetic operations. The other two pins (S1 and S0) are used to control the input and perform a variety of combinations. The various combinations of selection lines generate various operations, which are shown in Table V.

TABLE V. CONTROL OPERATION

S3	S2	S1	S0	Cin	Equation	Operation Name
0	0	0	0	X	$A \cdot B$	AND
0	0	0	1	X	$\bar{A} \cdot B$	$\bar{A}$ AND B
0	0	1	0	X	$A \cdot \bar{B}$	A AND $\bar{B}$
0	0	1	1	X	$\bar{A} + \bar{B}$	NOR
0	1	0	0	X	$A + B$	OR
0	1	0	1	X	$\bar{A} + B$	$\bar{A}$ OR B
0	1	1	0	X	$A + \bar{B}$	A OR $\bar{B}$
0	1	1	1	X	$\bar{A} \cdot \bar{B}$	NAND
1	0	0	0	X	$A \oplus B$	XOR
1	0	0	1	X	$\bar{A} \oplus B$	XNOR
1	0	1	0	X	$A \oplus \bar{B}$	XNOR
1	0	1	1	X	$\bar{A} \oplus \bar{B}$	XOR
1	1	0	0	0	$A + B$	Simple Add
1	1	0	0	1	$A + B + \text{Cin}$	Add with carry
1	1	0	1	0	$A + \bar{B}$	Subtract with borrow
1	1	0	1	1	$A + \bar{B} + 1$	Full subtract
1	1	1	0	0	$A - 1$	Decrement
1	1	1	0	1	A	Pass A
1	1	1	1	0	A	Pass A
1	1	1	1	1	$A + 1$	Increment

IV. PERFORMANCE EVALUATION

This section presents the simulation results for three types of ALUs: the Conventional ALU, the Conventional Reversible ALU, and the Proposed Reversible Gate-Based ALU. The simulations were executed utilizing Cadence Virtuoso software, employing 16nm CMOS technology, with performance evaluation based on power, delay, and energy metrics. Figure 14 and 15 demonstrates the input/output waveform of proposed reversible gate-based ALU. Figure 14 represents the first part of seven different arithmetic operations, such as addition, subtraction, increment, decrement, add with carry, 2's complement subtraction, and transfer voltage. On the other hand, Figure 15 mentions six different types of logical operations, including AND, OR, XOR, XNOR, NOR, and NAND gates.

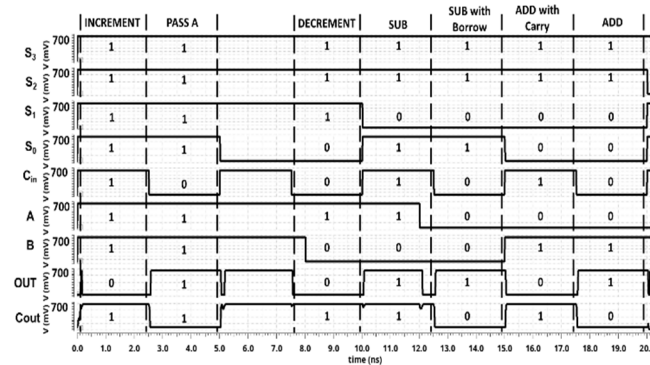


Fig. 14. Simulated output for proposed reversible ALU Arithmetic functions.

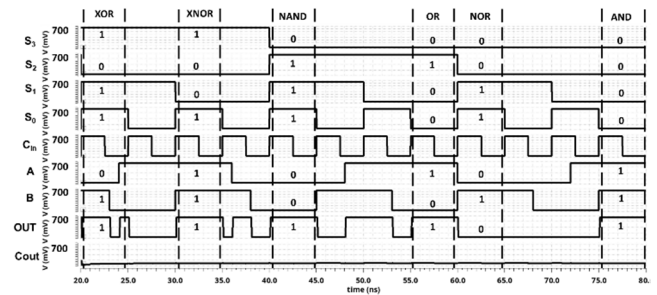


Fig. 15. Demonstration of logical operations for proposed reversible ALU.

Power, delay, and energy are essential for the analysis and optimization of digital circuits, especially in low-power applications. Power is the rate of energy consumption, encompassing dynamic power from switching and static power from leakage currents [12]. Figure 16 displays the power usage of various 8-bit ALUs, which includes the conventional 8-bit CMOS-based ALU, the conventional 8-bit reversible gate-based ALU, and the new 8-bit reversible gate-based ALU. Additionally, the proposed ALU's power consumption is significantly lower at 18.91 μW , which represents improvements of 69.74% and 69.38% compared to the existing 8-bit CMOS-based ALU and the existing 8-bit reversible gate-based ALU, respectively.

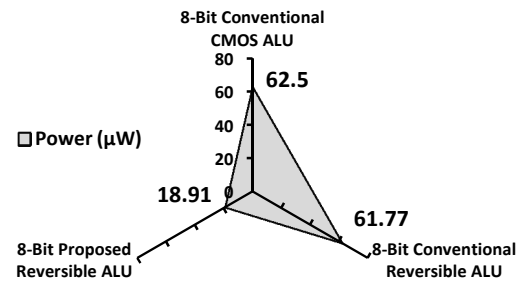


Fig. 16. Comparison of power dissipation between proposed and conventional ALUs.

The concept of "delay" refers to the amount of time it takes for a signal to propagate traveling the input to the output in a device [13]. Figure 17 illustrates the delay analysis of an 8-bit CMOS-based and reversible gate-based ALU to the proposed 8-bit reversible gate-based ALU. The proposed ALU provides delay at 61.86 pS, which is lower than other existing ALUs. The CMOS-based ALU has the worst delay at 83.6 pS. Overall, the proposed ALU delays are 26.6% and 21% quicker than conventional CMOS and reversible gate-based ALUs, respectively.

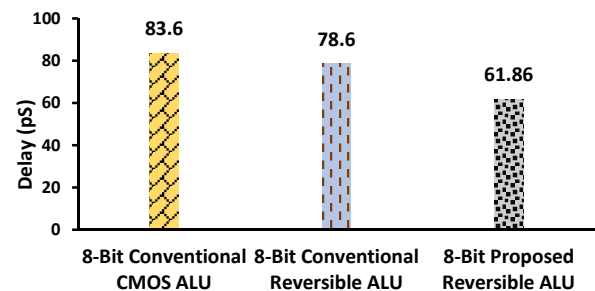


Fig. 17. Comparison of delay between proposed and conventional ALUs.

Energy consumption is the amount of energy needed to operate a circuit and complete perform logic operations; it tends to be measured employing the dynamic, static, and short-circuit power components throughout a certain period of time [13]. The development of efficient digital systems is based upon the specified metrics. Energy consumption of existing 8-bit ALUs and the proposed reversible-based 8-bit ALU is illustrated in Figure 18. Conventional CMOS-based ALUs absorb the most energy. The suggested ALU uses 1.51 pJ less energy and is 67.87% and 66.38% more efficient than the 8-bit conventional CMOS-based ALU and the 8-bit conventional reversible-based ALU, respectively.

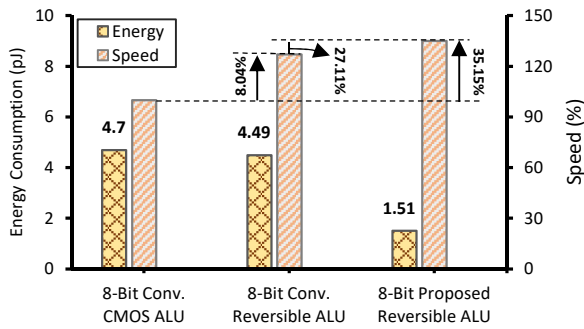


Fig. 18. Energy consumption and speed analysis of the differences between the proposed and existing ALUs.

Table VI compares how well the 8-bit proposed reversible gate-based ALU performs against [14] & [15] reversible gate-based ALUs and [4], [12] & [16] traditional CMOS-based ALUs. The proposed ALU consumes only 0.01891 mW, which is a significant reduction in power consumption of 99.95%, 95.02%, 27.26%, 93.9%, and 99.93% compared to [14], [15], [4], [12], and [16], respectively. As opposed to, the suggested ALU is 219.5 $\times$, 675.7 $\times$, 186.4 $\times$, and 1711.9 $\times$ faster than [15], [4], [12], and [16], respectively. This result illustrates that the proposed design is well-suited for low-power applications.

TABLE VI. COMPARISON BETWEEN EXISTING AND PROPOSED ALU

Parameters	[14]	[15]	[4]	[12]	[16]	Proposed
Power (mW)	38.75	0.38	0.026	0.310	28.02	0.01891
Delay(ns)	0.005	13.58	41.8	11.53	105.9	0.06186

V. CONCLUSION

This study demonstrates reversible ALU design that is optimized for efficient performance and minimal power consumption, utilizing Feynman, Fredkin, Peres and URG gates. The circuit was developed and simulated using Cadence Virtuoso 16nm technology to assess power, delay, and energy parameters. The gate-level control unit is the primary feature of this ALU. This control mechanism activates a single operational unit at any given moment, allowing the design to achieve approximately 69.74% and 69.38% improvements in power consumption, 26% and 21.29% enhancements in speed, and 67.87% and 66.38% decreases in energy usage relative to conventional CMOS and reversible ALU architectures, respectively. The results emphasize the suitability of the proposed ALU for quantum computing and low-power applications.

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